

Intra-household Bargaining, Endogenous Divorce, and Female Labor Supply

By TIANZHOU YAO*

This paper studies how intra-household bargaining shapes female labor supply and marital dynamics. I develop a heterogeneous-agent life-cycle model with a two-earner collective framework and endogenous divorce, incorporating education-dependent marital preferences. Calibrated to Japanese panel data, the model replicates substantial heterogeneity in labor supply, savings, and divorce across household types. To provide causal evidence, I exploit the 2007 introduction of Japan's Employee Pension Division System, which increases women's expected post-divorce pension wealth. Using a difference-in-differences design based on pre-reform income imbalance, I estimate how shifts in outside options affect behavior. The results highlight that policies altering long-term resource distribution within households can have immediate effects on labor supply through bargaining channels. More broadly, the findings suggest that aggregate labor market trends partly reflect changes in household composition rather than uniform behavioral responses.

This paper examines the effect of intra-household bargaining on extensive female labor supply decisions. Allowing for asymmetric marital preference shocks within a married couple, their "tastes for marriage" depend on their gender and education level. Divorce is an outside option that will be chosen endogenously when the expected life-time value of a divorcee is larger than the expected life-time value of remaining in the marriage. Relaxing the unitary model assumption in most life-cycle dynamic models, I construct a structural model that controls for all the well-established determinants of increasing female labor supply, and focus on particularly the role of cross-sectional variation of female bargaining power within a household.

To understand why does intra-household bargaining matter for female labor force participation decisions, I start from analyzing a simple non-unitary model with only two periods young vs. old. The analytical results suggest that private consumption allocation is proportional to the Pareto weight. The "working premium" in terms of expected life time value is inversely proportional to the wife's bargaining power. However, if I allow for elastic working hours decision, female will choose to work if only they are given a high enough bargaining power inside the household.

Starting from here, I am particularly interested in under both earnings shock

* Department of Economics, University of Washington, 410 Spokane Lane, Savery Hall, Seattle, WA 98195-3330 (e-mail: tianzyao@uw.edu).

and marital preference shock, how will different types of households by education levels and their asset levels affect consumption decision and female labor force participation decision. The model indicates that life-cycle profile of asset accumulation for married couples increases with their age until retirement. Households with non-college educated husbands tend to accumulate more asset at earlier stage but the trend is reversed when they get older. Related to the life-cycle profile of asset is the life-cycle profile of consumption, the model result shows inverse-u shaped consumption pattern where married couples consume the most in the middle of their working age. There is significant difference in consumption levels by the types of husband. A main focus of this paper is the female labor force participation pattern. Married females have approximately u-shaped participation patterns throughout their life-cycles, while non-college educated married females experience more prominent decreasing participation as they age leading to lower rate of participation than their college-educated counterparts in their mid 50s. Controlling for the education levels of married females, households with college-educated husbands have lower female labor force participation than households with non-college-educated husbands.

To uncover the parameters of the life-cycle structural model with four household types by education levels, I match the simulated model results with empirical targets derived from Japanese Panel Survey of Consumption (JPSC). The wage processes match corresponding average hourly wages for four types of individuals by gender and education levels. Another exogenously calibrated key parameter would be the Pareto weight for each type of household and for each asset level. To be consistent with the estimated result from Lise and Yamada (2019), I normalize the distribution of the Pareto weight so that the mean Pareto weight on wife is 0.44, with a maximum of 0.70 and a minimum of 0.20. Fixed utility cost of male is normalized to be zero since the model assumes full participation of married male, consistent with the data. Fixed utility costs of female participation are endogenously calibrated to match the mean female participation rates in 4 types of households respectively. A group of marital preferences parameters in the model including standard deviation of match quality shocks by genders and marital compatibility by combination of education levels are calibrated endogenously to match the probability of getting divorced. These estimates are embedded in the hypothesis of mutual consent divorce combining with community property regimes in which marital assets and debts are divided equally between the spouses upon divorce.

The importance of marital sorting by education to female labor force participation can be explained by the intra-household bargaining process. Positive assortative matching is most common in the sample, however, females in those households also have lowest bargaining powers, leading to lower participation. Negative assortative matching consists only 1.5% of the sample, has significantly higher female labor force participation because females have the highest bargaining power. Female participation rates are negatively correlated with asset levels

for all types of households, indicating negative "income effect". The model also shows that the probability of divorce relies largely on the education level of wives, meaning households with college-educated females see significantly larger chances of divorce compared with their non-college-educated counterparts.

To provide empirical discipline and policy relevance, this paper complements the structural analysis with quasi-experimental evidence from the 2007 introduction of the Employee Pension Division System in Japan. The reform allows divorcing couples to split earnings-related pension records accumulated during marriage, thereby increasing the expected post-divorce pension wealth of the lower-earning spouse. In the context of the model, this policy constitutes an exogenous shift in the wife's outside option and, consequently, her intra-household bargaining power. I exploit this reform using a difference-in-differences design that leverages cross-sectional variation in pre-reform income imbalance across households. The empirical results indicate that policies altering long-term resource allocation within households can generate immediate behavioral responses in labor supply and marital decisions. This evidence provides external validation for the model's central mechanism and highlights the broader policy implication that reforms targeting retirement wealth distribution may have unintended but important effects on labor market outcomes and family dynamics.

The contribution of this paper is threefold. First, I develop and estimate a dynamic model that explicitly incorporates varying Pareto weights at the time of the marriage. Voena (2015) recognizes the fact that under unilateral divorce, favorable intra-household bargaining position for wives provides a straightforward explanation for the reduction in their likelihood of employment that is observed in survey data. However, her paper lacks the heterogeneous agent dimension by assuming identical households at the time of marriage.

Second, I allow for endogenous marital transition and apply it to estimating life-cycle moments. Lise and Yamada (2019) interpret the Japanese Panel Survey of Consumption data using a dynamic collective model of household with potentially limited commitment. Taking into account the cross-sectional differences at the time of marriage in expected wage profiles between a husband and wife, they find that it strongly affects the allocation of private consumption by households. Their paper not only lacks the life cycle profiles critical for analysis of asset accumulation, consumption, and female labor force participation, but also lacks the endogenous family composition transition. Krueger and Wu (2021) show that a calibrated life-cycle two-earner household model with endogenous labor supply can rationalize the extent of consumption insurance against shocks to male and female wages. They find that most of the consumption insurance against permanent male wage shocks is provided through the presence and labor supply response of the female earner. Since they base their analysis on the unitary model, the relative bargaining power inside a household and its cross-sectional variation are missing from the factors that would possibly affect the insurance channel against income shocks.

Third, I generate asymmetric marital preferences within a couple. Divorce is either modeled as an outside option of intra-household bargaining in homogeneous households (Mazzocco 2007; Mazzocco, Ruiz, and Yamaguchi 2007; Voena 2015), or it is modeled as an exogenous shock generated by a stochastic process (Borella et al. 2023; Cubeddu and Rios-Rull 2003; Sander 1985). Empirical observations indicate that fraction of ever married men age 35-39 increases with education level, while fraction of ever married women age 35-39 decreases with education level. Rather than leaving it to the residual, my paper explicitly models the taste for marriage and assumes that it varies with gender and education level. Aside from income shocks, the risk of divorce followed by decreasing assets and consumption would be another channel unique to my paper that affects the female labor force participation decision. Calibrated to Japanese panel survey data, it is important to account for the background empirical trends including increasing divorce rates and decreasing marriage rates that change the demographic composition of the population.

I. Literature Review

Rising female labor force participation rate starting in the twentieth century in the US was named "Quiet revolution" by Goldin 2006. A contentious literature has attempted to establish its driving forces and implications. This important trend is accompanied by increasing family composition shocks including rising divorce rate and declining marriage rates, especially prominent in East Asian countries. The literature suggests positive correlation between this demographic change and rising female labor force participation, though less from an intra-household bargaining point of view. To address the issue, a small but growing literature focus on macro-implications of intra-household bargaining that brings new light to measuring income inequality and wealth inequality.

A. *Rising Female Labor Force Participation*

Women's increased involvement in the economy was the most significant change in labor markets in the past century. An important strand of literatures studies the determinants of rising female labor force participation in the 20th century.

Heathcote et al. (2009) features all the key forces behind the increase in female participation: 1) the decline in marriage rates, 2) the narrowing gender wage gap, 3) the preference (or cultural) shift towards market work, and 4) the change in women's bargaining power within the household. They find that preference shifts and the rise in relative wages of women were the most important driving forces behind rising women's participation. Attanasio et al. (2008) focus on the increase in labor supply of mothers between 1940s and 1950s cohorts in the US. They construct a life-cycle model of female participation and savings and investigate which changes in the determinants of labor supply account for the increases in participation early in the life cycle. They find that a combination of a reduction

in the cost of children alongside a reduction in the wage-gender gap is needed. Albanesi and Olivetti (2016) argue in their paper that medical progress played a critical role in the process of married women entering the labor force. They construct economic measures of improved maternal health and introduction of infant formula as two dimensions of medical progress and find that these advances indeed help women to reconcile work and motherhood.

My research contributes to the existing literature by 1) relaxing the unitary theoretical framework that allows bargaining power to vary cross-sectionally with distribution factors such as household wealth and spousal wage gap, 2) relaxing the full commitment assumption and model bargaining power as an endogenous variable that characterize renegotiation inside the household after new information regarding earnings (or preferences) are realized.

B. Increasing Family Composition Shocks

In the US, the introduction of unilateral divorce law in states that imposed an equal division of property is associated with higher household savings and lower female employment, implying a distortion in household assets accumulation and a transfer toward wives whose share in household resources is smaller than the one of their husbands. As discovered by Voena (2015), how divorce laws affect couples' intertemporal choices and well-being can be uncovered through explicit modelling of intra-household bargaining process that captures the relationship between higher risk of divorce and household behavior.

In Spain, over the past twenty years there has been an 87% increase of married women's employment rate and a 48% decrease of the fertility rate. Sanchez-Marcos (2002) focus on the increase of the probability of divorce from 0.5% to 2.1% following the introduction of the Divorce Law, as a possible explanation. The paper finds that the increase in marital dissolution rate can explain up to 44% of the change in married women's full-time employment rate, depending on women's education. The model on the economics of the family, in which labor market and fertility decisions are made, provides a good framework for the evaluation of certain social policies that target the reconciliation of family and labor life.

Cubeddu and Rios-Rull (2003) show in their paper the quantitative importance of the process that determines changes in family composition to the main macroeconomics magnitudes. They model family type as a stochastic process that affects households in a way like shocks to earnings. Agents respond to these processes by optimally choosing savings. They show that the size of savings differs dramatically depending on the details of the stochastic process.

My research contributes to the existing literature by 1) constructing gender-education-specific probability of divorce induced by heterogeneous distribution of marital preference shocks, 2) allowing for endogenous marital transition when the participation constraint under limited commitment framework binds, meaning divorce is an outside option when the expected lifetime utility of marriage is

smaller than that of divorce.

C. Modeling the Inter-temporal Behavior of Households

There is a small, but growing, literature that opens the family black box within macro models. Over the last few decades the literature on intra-household allocations has moved away from the unitary approach of Becker (1973, 1974, 1981) to models of strategic interaction between household members. Cooperative (Manser and Brown, 1980; McElroy and Horney, 1981) and non-cooperative bargaining solutions (Lundberg and Pollak, 1993; Del Boca and Flinn, 2012; Doepke and Tertilt, 2017) are common among models of intra-household interactions. However, the need to clearly specify the outside option, or threat point, of a marriage in these models poses difficulty for empirical analysis.

The collective model of household labor supply is an approach developed to circumvent this threat point issue (Chiappori, 1988, 1992; Apps and Rees, 1988). Without identifying the exact bargaining process within a household and the outside option of spouses, they assume households naturally arrive at Pareto efficient allocations. The important finding concludes that based on observant (leisure, private consumption or labor force participation), we can assign different individual bargaining power characterized by Pareto weight to the husband and wife that shows the sharing rule of total household resources.

Substantial cross-sectional empirical studies demonstrate that allocations within households are related to factors such as types of income, marriage market attributes like sex ratio and legislation that determines division of property upon divorce. Attanasio and Lechene (2014) provide compelling evidence rejecting the unitary model of the household in favour of a collective model. They use the exogenous variation in the wife's share of household income induced by a cash transfer programme in Mexico to implement the test of the collective model. Mazzocco (2007), on the other hand, highlights the substantial issues raised by the degree of commitment. He proposes a test for full commitment (ex ante efficiency), against no commitment (a series of static interactions) or limited commitment (limited only by participation constraints). However, the data used does not allow measuring separately the private consumption of the husband and the wife and does not contain a panel dimension. As a result, it is not possible to estimate the relative importance of cross-sectional dispersion in weights from changes accruing during marriage.

We see a lot of potential for using tools that are widely used in macroeconomics to build improved models of family. In particular, almost all macroeconomic models are dynamic, whereas in family economics static modeling is still common. There is a small literature that documents the importance of dynamics for the family. In particular, Mazzocco (2008) shows empirically that Euler equations hold at the individual but not the household level, Lise and Yamada (2015) provide evidence suggesting that bargaining power within the household evolves over time. To capture such phenomena and to better understand the link be-

tween family decisions and aggregate outcomes, more dynamic family bargaining models are needed.

Recently, Voena (2015) studied the effect of changes in divorce laws across the U.S. states (from mutual consent to unilateral), combined with different rules as to how assets are allocated at divorce (title based or equal division), on the savings and labor supply behavior of households. She finds that the observed responses of household asset accumulation and the labour supply response of the wife support a model of limited commitment in households. This analysis, however, assumes that the household weight at time of marriage is identical for all households.

My research contributes to the existing literature by focusing on life-cycle profiles of asset accumulation, consumption, and female labor force participation.

II. Motivating Facts

I first document empirical patterns indicating that marital preference is closely associated with probability of divorce by household types, and how does the event of divorce affect post-divorce female labor market outcomes. First, the average marginal effect of the unsatisfactory marital relationship reported by the wife results in an increase in the probability of divorce. Second, when I examine female labor force participation decision around the event of divorce, those married to college-educated males display a significant dip in post-divorce participation. They also fail to recover pre-divorce participation level.

This study utilizes the longitudinal dataset from Japanese Panel Survey of Consumers (JPSC). JPSC is a survey of women consisting of 5 cohorts, and it is an ongoing annual panel dataset that has been collected since 1993. The data provides comprehensive information on household members and their individual characteristics, allowing for the analysis of joint spousal characteristics. The longitudinal nature of the dataset enables the examinations of changes the households experience over time.

For the analysis, I construct an event history for each respondent covering employment, wages, education level, marital status, marital satisfaction, and child presence. Annual employment and wage information for all household members allows me to study labor market decisions and outcomes for both spouses in a joint household context. JPSC also collects marital satisfaction of wives as a variable that reflects marital preferences. When asked "Are you satisfied with your relationship with your husband?", wives chose one of the following responses that best matches their level of marital satisfaction: 1. Very much 2. Quite a lot 3. Moderately so 4. A little 5. Not at all. Specifically, I define "Marital unsatisfactory" as wives' response being either 4 or 5. The reason for using this variable is to understand the effect of individual marital preference on divorce, taking into account that under mutual consent divorce regime there is no way to infer which party initiates the motion. Finally, JPSC contains detailed information on all household members, including children, which allows me to track basic characteristics of each child.

Because I am interested in family composition transitions of working-age married women, the main sample includes married couples in which both spouses are aged 25-54, are observed as a couple in the current and prior waves, and have complete information on education, employment, earnings, and child presence. I restrict to waves 1993-2013. Table 1 presents average unsatisfactory rates across waves. In my sample, females with college degrees who were married to males without college education have the lowest unsatisfactory rate, on the contrary, those with the same education level but have college-educated husbands have the highest unsatisfactory rate among all types of households. The descriptive statistics indicates heterogeneous marital preferences by gender and education level. Table 2 presents female labor force participation rates by marital status and married couple type. Consistent with prior work, single women are more likely to work than married women with 85% of participation rate. Additionally, females without college degree whose husbands had college education have the lowest participation rate, only around 50%. This indicates female labor force participation is not only associated with one's own education level, but is also associated with the education level of the spouse.

Table 1—: Marital Unsatisfactory Rate by Household Type

Household Type	Unsatisfactory Rate
LL	0.122
LH	0.045
HL	0.178
HH	0.225
Total	0.124

Table 2—: Female Labor Force Participation by Household Type

Household Type	Participation Rate	N
LL	0.623	1,663
LH	0.537	41
HL	0.511	262
HH	0.682	66
Single	0.849	1,805
Total	0.722	3,837

Note: Household types are assigned based on education level of the spouse, where L means no college education, H means at least 4 years of college education.

A. Female Marital Satisfaction and Divorce

I begin by documenting how wives' marital preference is associated with divorce probability, focusing on educational differences between types of households. In this section, I show that the average marginal effect of marital preferences differs by types of households. Couples with the same educational level are more sensitive to marital unsatisfactory, meaning that, 1% of increase in marital unsatisfactory leads to 3 to 5 pp of increase in divorce probability. I document these patterns in a logit estimation framework, and employ a regression as follows:

$$\begin{aligned} Divorced_{i,t+1} = & \alpha + \beta UnsatW_t + \sum_k \gamma_k^B \mathbf{1}\{Type_i = k\} \\ & + \sum_k \delta_k^B UnsatW_t \mathbf{1}\{Type_i = k\} + X'_{i,t} \theta + \lambda_t + u_{i,t}, \end{aligned}$$

where $Divorced_{i,t+1}$ represents the outcome variable of interest, which is a dummy representing divorce for individual i in year $t + 1$. The equation regresses the outcome variable on a dummy variable $UnsatW_t$ which equals to 1 if the wife is unsatisfied with her marriage, types of households dummies (excluding base type LL), and the interaction variable between household types and marital unsatisfactory. $X_{i,t}$ is a vector that includes control variables like duration of marriage, age of spouses, number of children between age 0 and 6, the presence of newborn, and labor market participation of spouses. I employ year fixed-effects in λ_t . The age indicators capture life-cycle trends, while the calendar year indicators account for broader time trends. Using this logit estimation, I analyze how female marital preference is associated with changes in family composition across heterogeneous households. The marginal effects of marital unsatisfactory on divorce probability by household types is shown in Figure 1. Higher marital unsatisfactory is associated with larger probability of divorce across all types, while the magnitude of the effect varies by types of households.

B. Female Labor Force Participation around Divorce

I begin by documenting how divorce is associated with labor market outcomes of women. I show that prior to divorce, women's labor force participation is mostly associated with their own type. After divorce, labor force participation patterns vary by their husbands' type, with women who married H type men experiencing large dips in participation after divorce. I document these patterns in an event-study framework, and employ an event-study specification expressed as follows:

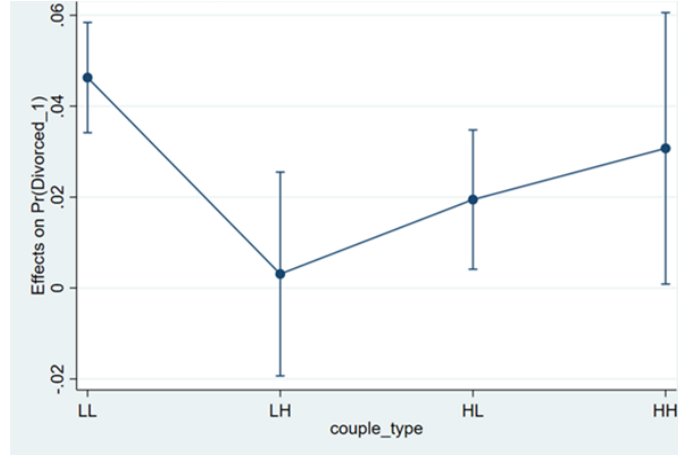


Figure 1. : Marginal Effects of Marital Unsatisfactory on Divorce Probability

$$\begin{aligned}
 pw_{i,t} = & \alpha + \sum_{\ell \neq -1} \beta_{\ell} \mathbf{1}\{event_{i,t} = \ell\} + \sum_k \gamma_k \mathbf{1}\{Type_i = k\} \\
 & + \sum_{\ell \neq -1} \sum_k \delta_{\ell k} \mathbf{1}\{event_{i,t} = \ell\} \mathbf{1}\{Type_i = k\} + X'_{i,t} \theta + \lambda_t + u_{i,t},
 \end{aligned}$$

where $pw_{i,t}$ represents female participation for individual i at event time t . The event time t is indexed relative to one year before the divorce. The equation regresses the outcome variable on event time dummies, controlling for age and calendar year indicators. Additionally, I control for the household types, the interactions between the type dummies and event time dummies. I also include other controls in the vector $X_{i,t}$ such as age of the wife, number of children between age 0 and 6, and the presence of newborn. Lastly, I employ year fixed effects λ_t , and standard errors are clustered at the couple level. The event time dummy at $t = -1$ is omitted, meaning that the event time coefficients β_{ℓ} represent deviations from the baseline period. Using this event study design, I analyze how the event of divorce is associated with changes in female labor force participation. Leveraging annual panel data on employment, I examine longer-term trends following divorce.

I examine post-divorce labor market outcomes for women from different types of households. Figure 2 shows that, controlling for wives' types, post-divorce labor market outcomes vary substantially depending on husbands' types. For those who were married to L type husbands, labor force participation increases after divorce. Especially for H type wives, the immediate increase of participation is predicted to be as large as 50%. However, for those who were married to H type husbands, labor force participation plummeted to around 0 immediately after

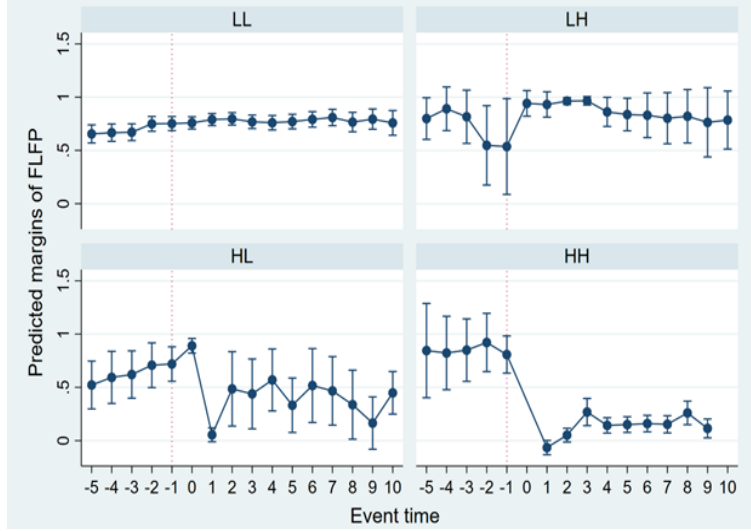


Figure 2. : FLFP around Divorce

divorce regardless of their own types.

Overall, these findings reveal patterns in divorce outcomes across household types that differ by educational combinations. It's also worth mentioning the heterogeneous post-divorce labor market outcomes for women that largely depend on the husband's type. Together, they suggest the need to open the family black-box, to introduce intra-household bargaining, marital preference, and family composition transition to the classic life-cycle structural model.

III. The Model

To identify the channels through which intra-household bargaining process affect consumption and female labor force participation decision, I first develop a simple non-unitary two-period dynamic model in which spouses jointly decide how to allocate consumption, and whether or not to work for the wife.

I then extend the simple model to a life-cycle model combined with the collective model of labor supply, in which spouses make the same decisions as in the simple two-period model in addition to the decision to get divorced. Both models take household formation as given, and hence is meant to examine the behaviors of couples that are already married.

A. A Simple Non-unitary Model

Two individuals, husband h and wife w , are married at time 0 and live until time 2. In each of the two periods, they jointly choose how much to save, how to allocate private consumption between one another. The household also makes

decisions about the wife's labor market participation. Husbands always work until they die. From time 1 to 2, spouses are retired. The working hours of spouses are set to be inelastic and normalized to one.

A.1 PREFERENCES

Both husband and wife derive utility from own consumption c_{jt} and the wife derive disutility from own labor force participation p_w for $j = h, w$ and $t = 1, 2$. θ is the Pareto weight on the husband.

Expected life time utility takes the form

$$\begin{aligned} & \theta \ln c_{h1} + (1 - \theta) (\ln c_{w1} - p_w \cdot \psi) \\ & + \beta [\theta \ln c_{h2} + (1 - \theta) \ln c_{w2}] \end{aligned}$$

A.2 BUDGET CONSTRAINT

The general form of the budget constraint is

$$c_{h1} + c_{w1} + \frac{c_{h2}}{1+r} + \frac{c_{w2}}{1+r} = w_{h1} + w_{w1} + \frac{w_{h2}}{1+r} + \frac{w_{w2}}{1+r}$$

A.3 OPTIMAL CONSUMPTION

Private consumption allocation chosen to maximize expected life time utility to the husband and the wife are, respectively

$$\begin{aligned} c_{h1}^* &= \frac{\theta ((1+r)(w_{h1} + w_{w1}) + w_{h2} + w_{w2})}{(1+\beta)(1+r)} \\ c_{w1}^* &= \frac{(1-\theta)((1+r)(w_{h1} + w_{w1}) + w_{h2} + w_{w2})}{(1+\beta)(1+r)} \end{aligned}$$

It can be shown from the result that private consumption allocated to each spouse is proportional to their bargaining power inside the household. The spouse would consume the corresponding share of discounted total resources in their life time, represented by the Pareto weight θ for the husband and $1-\theta$ for the wife.

A.4 FEMALE PARTICIPATION DECISION

Let $(1+r)(w_{h1} + w_{w1}) + w_{h2} + w_{w2} = R$, which indicates the life time total resources earned by the couple conditioning on the wife chooses to participate in the labor force. Let $(1+r)w_{h1} + w_{h2} + w_{w2} = R'$, which shows the life time total resources earned by the couple conditioning on the wife chooses not to participate in the labor market.

Then, the maximized life time utility of the married couple conditioning on the wife chooses to work, and conditioning on she chooses not to work are, respectively

$$\begin{aligned}
V_w &= \theta \ln \left(\frac{\theta R}{(1+\beta)(1+r)} \right) + (1-\theta) \left(\ln \left(\frac{(1-\theta)R}{(1+\beta)(1+r)} \right) - \psi \right) \\
&\quad + \beta \left[\theta \ln \left(\frac{\beta \theta R}{1+\beta} \right) + (1-\theta) \ln \left(\frac{\beta(1-\theta)R}{1+\beta} \right) \right] \\
V_{nw} &= \theta \ln \left(\frac{\theta R'}{(1+\beta)(1+r)} \right) + (1-\theta) \ln \left(\frac{(1-\theta)R'}{(1+\beta)(1+r)} \right) \\
&\quad + \beta \left[\theta \ln \left(\frac{\beta \theta R'}{1+\beta} \right) + (1-\theta) \ln \left(\frac{\beta(1-\theta)R'}{1+\beta} \right) \right]
\end{aligned}$$

For the household to determine collectively whether the wife should participate in the labor market, the couple compares between V_w and V_{nw} . The difference between the two values can be expressed as

$$Diff = (1+\beta) \ln \frac{R}{R'} - (1-\theta)\psi$$

Given the wife's Pareto weight at the start of the marriage $1-\theta$, the critical value $1-\theta^*$ such that when $1-\theta < 1-\theta^*$, the expected life time value of choosing to work is larger than that of choosing not to work, can be shown as

$$1-\theta^* = \frac{(1+\beta) \ln \frac{R}{R'}}{\psi}$$

The critical value $1-\theta^*$ depends on three variables: β , $\frac{R}{R'}$, and ψ . The larger β is, the larger $1-\theta^*$ is. This shows that the more patient a household is, the more likely a wife would choose to work. The larger $\frac{R}{R'}$ is, the larger $1-\theta^*$ is. This shows that the larger the contribution of the wife's income to the total household resources, the more likely a wife would choose to work. Lastly, the smaller ψ is, the larger $1-\theta^*$ is. This shows that the less the wife dislikes working, the more likely she would choose to participate in the labor market.

B. The Life Cycle Model

Two individuals, husband A and wife B , are married at time 1 and live until time T . In every period from time 1 to T , they jointly choose how much to save, how to allocate private consumption between one another, and whether to stay together or divorce. Between time 1 and time $T-R$, the household also makes decisions about the wife's labor market participation. Husbands always work until they retire. The working hours of spouses are set to be inelastic and normalized to one. From time $T-R+1$ to T , spouses are retired.

There are four types of household by education types (e_A, e_B) . $e_j \in \{H, L\}$ for individuals with or without college degree, respectively.

B.1 PREFERENCES

Both husband and wife derive utility from own consumption c_t^j and the wife derive disutility from own labor force participation p_t for $j = A, B$. Preferences are separable across periods of time and states of the world.

Both the husband and the wife have a subjective taste for marriage parameter $\xi_t^{j,e}$ which evolves over time and differs by gender and education level. This parameter reflects the spouses' affection for one another and their attachment to marriage based on other idiosyncratic factors (e.g., fear of the social stigma associated with divorce, concerns about the well-being of the children). Taste for marriage parameter only applies to married couples, that is, when $m_t = 1$. $s^{j,e}$ is stigma cost of divorce that only applies to divorcees, when $m_t = 0$.

Period utilities take the form

$$u_t^j = u(c_t^j, p_t) + \xi_t^{j,e} m_t - s^{j,e} (1 - m_t)$$

The taste shocks follow a random walk stochastic process, which captures the persistence in the taste for the current marriage,

$$\xi_t^{j,e} = \xi_{t-1}^{j,e} + \epsilon_t,$$

where ϵ_t is distributed as $N(0, \sigma^2)$.

At the start of the life cycle when $t = 1$, $\xi_1^{j,e}$ can take five values for the wife and the husband with or without a college degree.

$$\xi_1^{j,L} \in \{-2\sigma_j, -\sigma_j, 0, \sigma_j, 2\sigma_j\},$$

where

$$\xi_1^{A,H} = \xi_1^{A,L} + \sigma_A$$

,

$$\xi_1^{B,H} = \xi_1^{B,L} - \sigma_B$$

.

The utility function $u(c_t^j, p_t)$ is constant relative risk aversion (CRRA) and is separable in consumption and participation in the labor market:

$$u(c_t^j, p_t) = \frac{(c_t^j)^{1-\gamma}}{1-\gamma} - q_{-e} p_t$$

, with $\gamma \geq 0$ and $q_{-e} > 0$.

B.2 INCOME OVER THE LIFE CYCLE

Each spouse's labor income ($y_t^{j,e}$ for $j = A, B$ and $e = H, L$) depends on her human capital $h_t^{j,e}$ and on her permanent income ω_t^j :

$$\ln(y_t^{j,e}) = \ln(h_t^{j,e}) + \omega_t^j$$

Spouses experience permanent income shocks, which follow a random walk process:

$$\omega_t^j = \omega_{t-1}^j + \zeta_t^j \quad \text{and} \quad \omega_1^j = \zeta_1^j,$$

in which ζ_t^j is i.i.d. as $N(0, \sigma_\zeta^2)$ and is uncorrelated between spouses.

Human capital is accumulated through age. The law of motion for each spouse's human capital $h_t^{j,e}$ is

$$\ln(h_t^{A,e}) = \alpha_{0,e} + \alpha_{1,e}t + \alpha_{2,e}t^2$$

$$\ln(h_t^{B,e}) = \kappa_{0,e} + \kappa_{1,e}t + \kappa_{2,e}t^2 + \kappa_{3,e}t^3$$

Men always work until they retire. At the end of period $T - R$, spouses retire and receive a share of their preretirement income in every subsequent period.

B.3 BUDGET CONSTRAINT

In marriage, the general form of the budget constraint is

$$\frac{a_{t+1}}{1+r} - a_t + c_t = y_t$$

B.4 MARITAL COMPATIBILITY

Each type of couple has a subjective feeling for their current marital status, which differs by the education levels of both the wife and the husband (e_A, e_B). This parameter reflects the spouses' affection for one another and their attachment to marriage based on other idiosyncratic factors (e.g., how well partners in a marriage align across various domains, such as values, communication styles, decision-making, and conflict resolution).

Marital compatibility of a couple with education levels (e_A, e_B) is

$$M(e_A, e_B) = \mu_0(1 - e_A)(1 - e_B) + \mu_1 * e_A e_B$$

If neither attained a college degree, $M(0, 0) = \mu_0$. If both attained college degrees, $M(1, 1) = \mu_1$. $M = 0$ for the other two types of households.

B.6 TIMELINE FOR SHOCKS AND DECISIONS

There are two types of family in the model, families of a married couple and families of divorcee. At $t = 1$, each individual draws with equal probability an education level e_j of being college educated or not. In every subsequent time

period between time t and time $t + 1$, divorcees update the permanent wage shock ω_t^j , and then decide how much to save, how much to consume based on the realized shock. Female divorcees also make labor force participation decisions while male divorcees always work until retirement.

In every subsequent time period between time t and time $t + 1$, married couples update the gender-education-specific marital preference shock $\xi_t^{j,e}$ and the permanent wage shock ω_t^j . Based on the realized shocks, they make divorce decisions after comparing their expected life time value of marriage with the expected life time value of the outside option. Conditioning on the divorce decision being made, each household, either remaining as married couple and make decisions collectively about savings, consumption, and female labor force participation, or make individual decisions as divorcees:

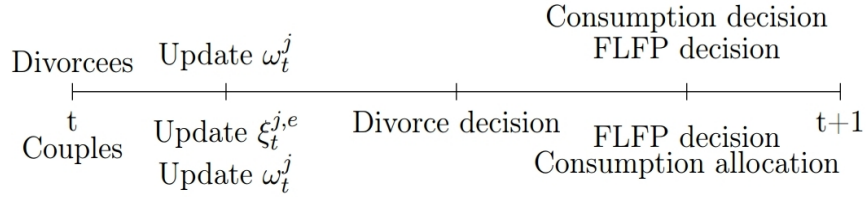


Figure 3. : Timeline for Shocks and Decisions

B.7 PROBLEM OF THE DIVORCEE

I now characterize the value of being divorced, given state variables $W_t = \{H, \omega_t^j, a_t^j\}$, where H represents the 2 types of divorcees by education level.

In each period t , the divorcee chooses consumption, savings, and whether or not to work (if she is a woman). Thus, the value of being divorced and not remarrying at time t is

$$V_t^{jD}(W_t) = \max_{p_t, a_{t+1}^j} u(c_t^j, p_t) + \beta E \left[V_{t+1}^{jD}(\omega_{t+1}^j | \omega_t^j) \right]$$

s.t. budget constraint in divorce, for $j = A, B$

$$\frac{a_{t+1}^j}{1+r} - a_t^j + c_t^j = y_t^j$$

The budget constraint depends on the property division regime at the time of divorce, which is assumed to be community property regime in the model

$$\frac{a_{t+1}^j}{1+r} - \frac{a_t^j}{2} + c_t^j = y_t^j$$

B.8 HOUSEHOLD PLANNING PROBLEM

The couple's planning problem depends on the current divorce law regime. In unilateral divorce, assumed by the model, the couple divorces even if just one spouse wants to divorce. The household solves a constrained Pareto problem, in which the husband's Pareto weight is indicated by θ_0 and the wife's weight by $1 - \theta_0$. The parameter θ_0 is determined exogenously.

Define $W_t = \{H, \theta_0, q_{-e}, M(e, e^*), \xi_t^{j,e}, \omega_t^j, a_t\}$ the vector of variables over which the household maximizes. A couple that enters period t as married solves

$$\begin{aligned} V_t^M(W_t) = & \max_{p_t, a_{t+1}} \theta_0 u^A(c_t^A) + (1 - \theta_0) u^B(c_t^B, p_t, q_{-e}) + M(e, e^*) \\ & + \beta E \left[V_{t+1}^M \left(\omega_{t+1}^j, \xi_{t+1}^{j,e}, \mid \omega_t^j, \xi_t^{j,e} \right) \right] \end{aligned}$$

s.t. budget constraint in marriage, for each married couple

$$\frac{a_{t+1}}{1+r} - a_t + c_t = y_t$$

B.9 DIVORCE DECISION

In a unilateral divorce state, the couple maximizes the weighted sum of spouses' utilities in marriage under the constraint that both spouses must prefer the marriage allocation to the value of being divorced. Which is to say, the marriage allocation has to satisfy the participation constraints by comparing the expected life time value of marriage V^{jM} with the expected life time value of divorce V^{jD} , for $j = A, B$.

There are three cases that will lead to divorce in the model

case 1: only wife wants to stay together, assume no renegotiation

$$V^{AM} \leq V^{AD} \quad \text{and} \quad V^{BM} \geq V^{BD}, \theta_{t+1} = \theta_t$$

case 2: only husband wants to stay together, assume no renegotiation

$$V^{AM} \geq V^{AD} \quad \text{and} \quad V^{BM} \leq V^{BD}, \theta_{t+1} = \theta_t$$

case 3: both want to divorce, no need for renegotiation

$$V^{AM} \leq V^{AD} \quad \text{and} \quad V^{BM} \leq V^{BD}, \theta_{t+1} = \theta_t$$

To align with the fact that in Japan, the predominant ground for divorce is mutual consent under the "divorce by mutual agreement", I assume accordingly that there will be no credible threat of divorce thus no adjustment of bargaining power within a household unless both of the spouses agree to separate. This indicated that the "limited commitment" channel is shut down under this divorce regime.

There is only one case that will leave to divorce in the benchmark model

$$V^{AM} \leq V^{A1D} \quad \text{and} \quad V^{BM} \leq V^{B1D}, \theta_{t+1} = \theta_t$$

C. Discussion of Mechanisms

This section develops the core implications of the model for (i) the effect of intra-household bargaining process on female labor force participation, (ii) how does asymmetric marital preference shocks between spouses reshape that effect. A formal derivation is in [Appendix Section](#).

C.1 COLLECTIVE LABOR SUPPLY UNDER WAGE SHOCK

Start from a symmetric static benchmark in which both spouses earn the same wage y_t . The optimal intra-period utility can be written in the form of

$$U_t^* = \theta_0 \frac{(S_A \cdot R)^{1-\gamma}}{1-\gamma} + (1-\theta_0) \left[\frac{(S_B \cdot R)^{1-\gamma}}{1-\gamma} - q_{-e} \right] + M(e, e^*)$$

where $S_A = \frac{\theta_0^{\frac{1}{\gamma}}}{\theta_0^{\frac{1}{\gamma}} + (1-\theta_0)^{\frac{1}{\gamma}}}$, and $S_B = 1 - S_A$. The net budgets $R = (1 - \tau_A) r y_t + (1 - \tau_B) y_t + a_t$ and $R' = (1 - \tau_A) \cdot r y_t + a_t$.

Consider a male only pay raise to $y_t^A = r y_t$ with $r > 1$, keeping wife's wage $y_t^B = y_t$ fixed. The cooperate gain from the wife dropping out of the labor market under the husband's wage increase can be written as

$$\Delta U_{y_t^B \rightarrow 0}(r) = \frac{(R(r)^{1-\gamma} - R(r)^{1-\gamma})}{1-\gamma} \left(\theta_0 S_A^{1-\gamma} + (1-\theta_0) S_B^{1-\gamma} \right) + (1-\theta_0) q_{-e}$$

where R' and R are the combined earnings after and before the wife drops out of the labor market. S_A and S_B are consumption shares of the husband and the wife respectively. The first term captures the effect on net income, and the second term captures the gain from avoiding disutility of working. Because $R(r)^{1-\gamma} - R(r)^{1-\gamma}$ is increasing in r and $(1-\theta_0) q_{-e}$ is positive, there exist finite threshold r^* such that the husband's wage raise induces the wife to drop out of the labor force if and only if r exceeds the relevant threshold [Appendix Section](#). For intermediate r , she remains full-time despite the higher male wage.

C.2 ASYMMETRIC MARITAL PREFERENCE SHOCKS

Assume before the marital preference shocks hit, both spouses have the same amount of additional utility ξ_t generated from marriage. After marital preference shocks hit, $(\xi_t^A, \xi_t^B) = (r \xi_t^A, \phi \xi_t^B)$ with $r > 1$ and $\phi < 1$. This means that I study the effect of changing marital preference where the husband derives more utility

from marriage and the wife derives less from marriage, hence the asymmetric marital preferences. If there is no adjustment in the Pareto weight, I can prove that the asymmetric marital preference shock has no effect on female labor force participation decisions since the cooperate gain from the wife dropping out of the labor market $\Delta U_{y_t^B \rightarrow 0}$ is not a function of r or ϕ . A formal proof is in [Appendix Section](#).

C.3 BARGAINING AND OUTSIDE OPTIONS

The previous sections analyzed the effects of imbalanced wage shock and asymmetric preference shocks under a given Pareto weight θ_0 . Here I describe how allowing for evolution of the Pareto weight feeds into the female labor force participation decision. Formal derivation of the cooperative problem and its comparative statics are provided in [Appendix Section](#).

The gap between cooperative and non-cooperative (divorce) utilities for each spouse can be summarized by the cooperation (marriage) surplus

$$P_t^j(\Omega_t; \theta_t) \equiv V_t^{jM}(\Omega_t; \theta_t) - V_t^{jD}(\Omega_t)$$

For both spouses to be willing to stay in the marriage, marital allocations must satisfy

$$P_t^j(\Omega_t; \theta_t) \geq 0, \quad j \in \{m, f\},$$

and deliver strictly positive joint surplus. Which participation constraint binds determines which spouse has leverage to renegotiate θ_t . When the wife's constraint binds, the Pareto weight is determined by the equation

$$V_t^{BM}(\Omega_t; \theta_t) = V_t^{BD}(\Omega_t).$$

The cooperation surplus for the husband can be written as

$$P_t^A = \frac{1}{1-\gamma} \left((S_{\hat{A}} \cdot R)^{1-\gamma} - (R_A)^{1-\gamma} \right) + r\xi_t,$$

and the cooperation surplus for the wife can be written as

$$P_t^B = \frac{1}{1-\gamma} \left((S_{\hat{B}} \cdot R)^{1-\gamma} - (R_B)^{1-\gamma} \right) + \phi\xi_t.$$

Because both P_t^A and P_t^B are increasing with r and ϕ , there exists a finite threshold $\frac{r}{\phi}^*$ such that when the degree of asymmetry of marital preference between spouses $\frac{r}{\phi}$ is larger than the threshold, $P_t^A > 0$ and $P_t^B < 0$ are satisfied simultaneously:

$$\left(\frac{r}{\phi} \right)^* = \frac{(S_A \cdot R)^{1-\gamma} - (R_A)^{1-\gamma}}{(S_B \cdot R)^{1-\gamma} - (R_B)^{1-\gamma}},$$

where R_j is the divorcee's total earnings. The wife's participation constraint binds, the updated Pareto weight $\theta_1 < \theta_0$ since the husband's bargaining power decreases.

Other shocks that affect the value of the outside option for the wife V_t^{BD} could also affect the Pareto weight when they cause the participation constraint to bind. The factors included in the full model are R_B and stigma cost of divorce G_B . I now show that whenever participation constraint binds, improvements in the wife's outside option such as increase in divorcee earnings (wages and assets) and decrease in stigma cost of divorce would translate into higher bargaining weight for her.

Define

$$F(\theta_t, V_t^{BD}) \equiv V_t^{BM}(\Omega_t; \theta_t) - V_t^{BD}(\Omega_t) = 0.$$

Because $V_t^{BM}(\Omega_t; \theta_t)$ is strictly decreasing in θ_t (see proof provided in [Appendix Section](#)), $\partial F / \partial \theta_t = \partial V_t^{BM} / \partial \theta_t < 0$ and $\partial F / \partial V_t^{BD} = -1$. By the implicit function theorem,

$$\frac{\partial \theta_t}{\partial V_t^{BD}} = - \frac{\partial F / \partial V_t^{BD}}{\partial F / \partial \theta_t} = - \frac{-1}{\partial V_t^{BM} / \partial \theta_t} = \frac{1}{\partial V_t^{BM} / \partial \theta_t} < 0.$$

Based on this result, I can also derive that $\partial \theta_t / \partial R_B < 0$, $\partial \theta_t / \partial G_B > 0$, $\partial \theta_t / \partial R < 0$ (see proof provided in [Appendix Section](#)). Aside from factors that directly improve the value of the outside option for the wife, an increase in total marital earnings could also increase her bargaining power. This aligns with various empirical findings in the literature that an increase in household asset level would make intra-household allocations more equal.

C.4 THE PARETO WEIGHT, THE GENDER MARITAL PREFERENCE GAP, AND FEMALE LABOR FORCE PARTICIPATION

This section revisits the labor force participation decision made by wife under asymmetric marital preference shock, but also allowing for adjustment of the Pareto weight. When the participation constraint binds for the wife, meaning that she is indifferent between staying in the marriage and divorce, the cooperative gain from the wife dropping out of the labor market under asymmetric marital preference shock can be written as

$$\begin{aligned} \Delta U_{y_t^B \rightarrow 0}(r, \phi) &= \xi_t(\theta_{t+1} - \theta_t)(r - \phi) \\ &\quad + E_t[\theta_{t+1}u(c_{t+1}^A; \theta_{t+1}) - \theta_t u(c_t^A; \theta_t) \mid \Omega_t] \\ &\quad + E_t[(1 - \theta_{t+1})u(c_{t+1}^B; \theta_{t+1}) - (1 - \theta_t)u(c_t^B; \theta_t) \mid \Omega_t] \\ &\quad + (1 - \theta_t)q_{-e} \end{aligned}$$

Define

$$E_t (\Delta V_{t+1}^A \mid \Omega_t) \equiv E_t [\theta_{t+1} u(c_{t+1}^A; \theta_{t+1}) - \theta_t u(c_t^A; \theta_t) \mid \Omega_t],$$

and

$$E_t (\Delta V_{t+1}^B \mid \Omega_t) \equiv E_t [(1 - \theta_{t+1}) u(c_{t+1}^B; \theta_{t+1}) - (1 - \theta_t) u(c_t^B; \theta_t) \mid \Omega_t].$$

Now the cooperative gain from wife dropping out of labor market becomes:

$$\Delta U_{y_t^B \rightarrow 0}(r, \phi) = \xi_t (\theta_{t+1} - \theta_t) (r - \phi) + E_t (\Delta V_{t+1}^A \mid \Omega_t) + E_t (\Delta V_{t+1}^B \mid \Omega_t)$$

Since $\theta_{t+1} < \theta_t$, and $R_{t+1} < R_t$, it implies $E_t (\Delta V_{t+1}^A \mid \Omega_t) < 0$. On the other hand, because the wife's participation constraint binds, she is worse off staying married comparing with the outside option which is divorce. After negotiation, the wife's bargaining power increases until she becomes indifferent between staying married and getting a divorce. This assumption results in two conditions that must be satisfied:

$$V^{BD} > V^{BM}(\theta_t) \text{ and } V^{BD} = V^{BM}(\theta_{t+1})$$

,

which lead to $E_t (\Delta V_{t+1}^B \mid \Omega_t) > 0$ (see detailed proof in [Appendix Section](#)).

When the participation constraint binds, $\Delta U_{y_t^B \rightarrow 0}(r, \phi)$ is strictly decreasing in $r - \phi$. Therefore, there exists a finite threshold $(r - \phi)^*$ such that when the gender marital preference gap between spouses $r - \phi$ is larger than the threshold, $\Delta U_{y_t^B \rightarrow 0}(r, \phi) < 0$ is satisfied. This implies when the wife is gaining bargaining power due to the improvement in her outside option, the more the husband prefers to stay married, the less the wife prefers marriage, the more likely it is for the wife to choose continue working full time.

The comparative statics show that the effect of the Pareto weight on female labor force participation depends on the gender marital preference gap:

$$\frac{\partial (\Delta U_{y_t^B \rightarrow 0})}{\partial \theta_{t+1}} = \xi_t (r - \phi) + \frac{\partial E_t (\Delta V_{t+1}^A \mid \Omega_t)}{\partial \theta_{t+1}} + \frac{\partial E_t (\Delta V_{t+1}^B \mid \Omega_t)}{\partial \theta_{t+1}}.$$

This mechanism can be seen as the bargaining channel that determines female labor force participation. Since $\frac{\partial E_t (\Delta V_{t+1}^A \mid \Omega_t)}{\partial \theta_{t+1}}$ is positive, $\frac{\partial E_t (\Delta V_{t+1}^B \mid \Omega_t)}{\partial \theta_{t+1}}$ is negative, it can be safely inferred that when the gender marital preference imbalance is large enough, that is, when $\xi_t (r - \phi) > \left| \frac{\partial E_t (\Delta V_{t+1}^B \mid \Omega_t)}{\partial \theta_{t+1}} \right|$, $\frac{\partial (\Delta U_{y_t^B \rightarrow 0})}{\partial \theta_{t+1}}$ will be strictly positive. This shows when the bargaining position of the wife gets better, the "non-working premium" will decrease, making it easier for the wife to choose to

stay working full-time.

The comparative statics with respect to the gender marital preference gap:

$$\frac{\partial \left(\Delta U_{y_t^B \rightarrow 0} \right)}{\partial (r - \phi)} = \xi_t (\theta_{t+1} - \theta_t) < 0.$$

Under the assumption that $\theta_{t+1} < \theta_t$, this derivative is strictly smaller than 0. When the gender marital preference gap widens, "non-working premium" for the wife decreases. This indicates the introduction of asymmetric marital preference shock makes it easier for the wife to reject dropping out of the labor force. This mechanism captures the direct effect of asymmetric marital preference shock on female labor force participation.

IV. Testing the Theoretical Implications: Policy Reform

A central implication of the model is that policies affecting spouses' outside options alter both marital stability and labor supply decisions. The 2007 introduction of the Employee Pension Division System upon Divorce in Japan provides a natural setting to test these mechanisms. The reform allows divorcing couples to divide the marital-period earnings-related pension records, thereby increasing the expected retirement resources of the lower-earning spouse—typically the wife.

This institutional change operates through two key channels. First, it reduces the economic cost of divorce for wives by granting access to a portion of the husband's pension wealth. Second, it alters intra-household bargaining power, as improved fallback options strengthen the wife's position within marriage. These channels generate testable predictions for both marital stability and labor supply.

To quantify the impact of the reform, I exploit its introduction as a quasi-exogenous policy change and implement a difference-in-differences (DID) design.

The analysis focuses on a panel of married women observed between 2005 and 2013, restricting the sample to those married prior to the reform (2005–2007). This ensures that all individuals in the sample are exposed to the policy change while avoiding selection from post-reform marriage formation.

Because the reform applies universally, identification relies on heterogeneous exposure intensity. I proxy exposure using pre-reform within-household income imbalance, measured by the wife's relative hourly wage averaged over 2005–2007. Couples in which the wife's relative wage falls below one-third are classified as more exposed, as the reform substantially increases their expected pension transfer upon divorce.

The baseline specification is:

$$P_{it} = \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + X'_{it} \gamma + \alpha_i + \lambda_t + \varepsilon_{it},$$

where P_{it} denotes the outcome of interest (female labor force participation), $Treat_i$ indicates low pre-reform relative wages, and $Post_t$ equals to one for years

after 2007. The model includes couple fixed effects (α_i) and year fixed effects (λ_t) and standard errors are clustered at couple level.

Table 3—: Effect of the Pension Division Reform on Female Labor Force Participation

	(1) <i>pw</i>
Treat $\times$ Post	0.1833*** (0.0175)
Age of wife	0.0608*** (0.0177)
Age of wife squared	-0.000492*** (0.000162)
Child aged 0–6	-0.0963*** (0.0120)
New child indicator	0.0108 (0.0345)
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.	

The estimates shown in table 3 indicate that the reform has a positive and statistically significant effect on female labor force participation among more exposed households. The coefficient on the interaction term is approximately 0.183, implying that, following the reform, women in highly imbalanced couples increase their labor force participation by 18.3 percentage points relative to less exposed couples.

This finding suggests that improvements in women’s expected pension rights translate into stronger labor market attachment. The result is consistent with a bargaining interpretation in which enhanced outside options increase women’s incentives to maintain or strengthen their labor market position.

The positive labor supply response highlights the importance of forward-looking behavior in household decision-making. Although pension benefits materialize only at retirement, the reform shifts expected lifetime resources and bargaining power immediately. As a result, women in more imbalanced households respond by increasing labor supply, potentially to reinforce their economic independence or to capitalize on improved bargaining positions.

At the same time, the magnitude of the effect suggests that the reform operates not only through retirement incentives but also through contemporaneous changes in household dynamics, including expectations about marital stability and intra-household allocation.

These findings provide empirical support for models in which policy-induced changes in outside options affect both specialization and bargaining margins. By increasing the value of the wife’s fallback option, the pension division system reshapes incentives within marriage, leading to observable changes in labor supply behavior.

More broadly, the results highlight that policies targeting long-term wealth distribution—such as pension rights—can have immediate behavioral effects, even in the absence of direct short-run transfers.

V. Model Estimation

In this section, I describe how I parameterize the model, using empirical targets derived from Japanese household data, as measured in the Japanese Panel Survey of Consumption (JPSC).

I use the JPSC data covering the period from 1993 to 2013. The data include four cohorts: cohort 1 comprising 1,500 women aged 24-34 years in 1993, cohort 2 comprising 500 women aged 24-27 years in 1997, cohort 3 comprising 836 women 24-29 years in 2003, and cohort 4 comprising 636 women aged 24-28 years in 2008. In addition to rich data on demographics, education, wages and labor supply, the JPSC has a consumption expenditure module and a time use module. I choose cohort 1 as the benchmark cohort. Table 4 reports summary statistics by household types of cohort 1. Figure 4 presents the comparison between cohort 1 and cohort 3 in terms of female participation and working hours. It shows higher female participation as well as higher female labor supply (conditional on working) everywhere in the life cycle for cohort 3. I keep married women and their husbands for whom I observe, for both spouses, (1) their demographic characteristics, such as current age, age at marriage, and education; (2) their expenditures and time use for at least two consecutive years; and (3) their wages for at least two consecutive years. The sample used in the baseline analysis includes 655 households (8,832 household-year observations).

The JPSC asks the following question about the components of household expenditures:

Please write down your household expenditure in September this year, including not only cash purchases, but also purchases with the credit loan(s), or those charged to your bank/post office account. (If there was no expenditure corresponding to the items below, put "0" for each answer.)

Importantly for my purpose, the JPSC also asks for breakdown of total household expenditures into the following five categories: (1) Expenses for all of your family, (2) Expenses for you, (3) Expenses for your husband, (4) Expenses for your child(ren), (5) Expenses for other(s).

I treat categories 1, 4, and 5 as expenditures used to produce the household public good, category 2 as private consumption of the wife c^B , and category 3 as the private consumption of the husband c^A .

Since the aim of this paper is to examine the life cycle patterns of consumption,

Table 4—: Summary Statistics by Household Type

	LL (63.2%)	LH (1.5%)	HL (26.1%)	HH (9.2%)
Panel A: Wife				
Expenditure	133,673 [2.91]	153,624 [3.84]	162,707 [3.02]	208,178 [4.29]
Age	37.3 [4.30]	39.7 [3.39]	37.8 [4.54]	37.8 [4.27]
Wage	775 [0.06]	870 [0.07]	856 [0.08]	1,054 [0.07]
Assets	336 [395.53]	762 [837.63]	557 [618.30]	840 [841.66]
Panel B: Husband				
Expenditure	311,579 [4.32]	252,938 [4.99]	334,565 [3.85]	328,225 [4.11]
Age	39.8 [5.74]	39.8 [4.29]	41.3 [5.26]	40.0 [5.87]
Wage	1,544 [0.07]	1,859 [0.09]	1,894 [0.07]	1,955 [0.06]

Note: Entries are means, with standard deviations in brackets. Household-type shares are reported in column headers.

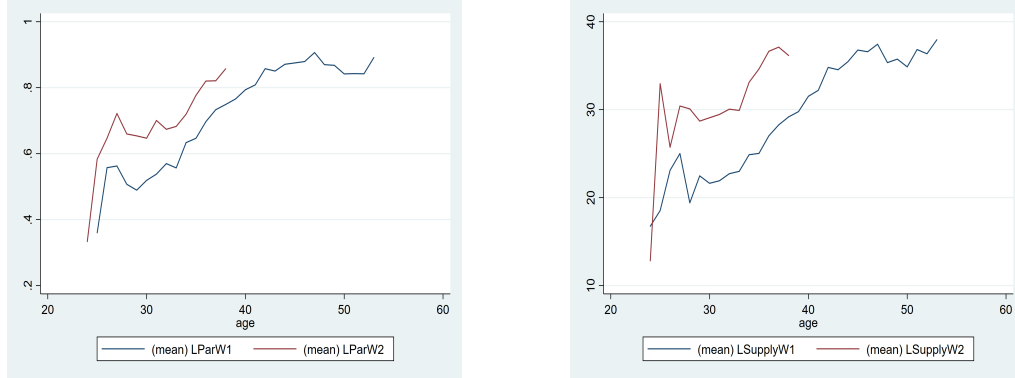


Figure 4. : Cohort Comparison of Female Labor Supply Pattern

asset accumulation, and female labor force participation, I therefore present in figure 5, 6 and 7 the life cycle patterns by household types observed in the JSPC data for the benchmark cohort. The left graphs show cross-sectional mean of consumption, asset and female labor force participation at each age generated from raw data for household types $e^A e^B$. To account for fixed effects within group and between groups, I run random effect regression to derive the predicted variables that filters out fluctuations:

$$v_{it} = \alpha + \beta * AgeW_{it} + \sum_{j=2}^4 \gamma_j D_{j,it} + \sum_{j=2}^4 \delta_j (AgeW_{it} \times D_{j,it}) + u_i + e_{it},$$

where v_{it} represents outcome variables including household consumption, savings, and female labor force participation. I regress the outcome variables on age of the wives, house-type dummies $D_{j,it}$, and their interaction. u_i accounts for the random effect. Results are shown on the right graphs of figure 5-figure 7 as predicted life-cycle profiles of consumption, asset accumulation, and female labor force participation.

A. Demographic and Initial Conditions

Households are born at age 21, when they start working. They retire at age 65, and die at age 80. Therefore, age 1 in the model corresponds to age 24 in the data, and consequently $t_R = 42$ and $T = 57$. Households start their life with assets ranging from 150 to 1700 that vary by household types and a permanent component of log hourly wages ranging from -2.308 to -1.821 based on individual's gender and education level. There is no transitory component for log wages.

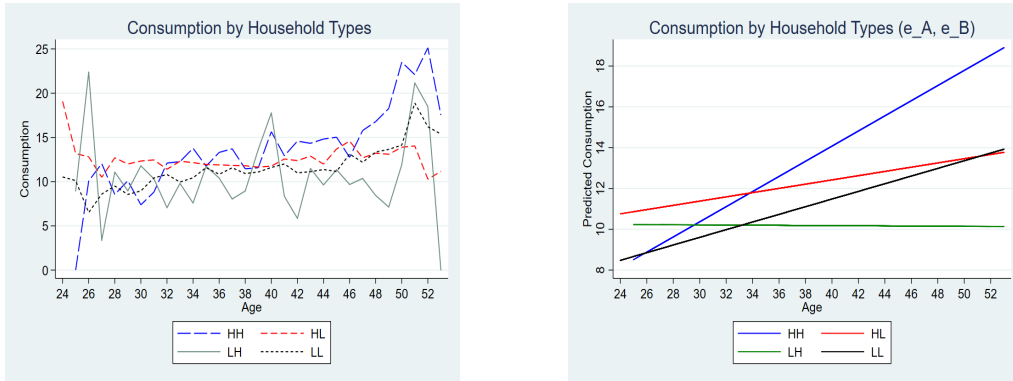


Figure 5. : Consumption by Household Types

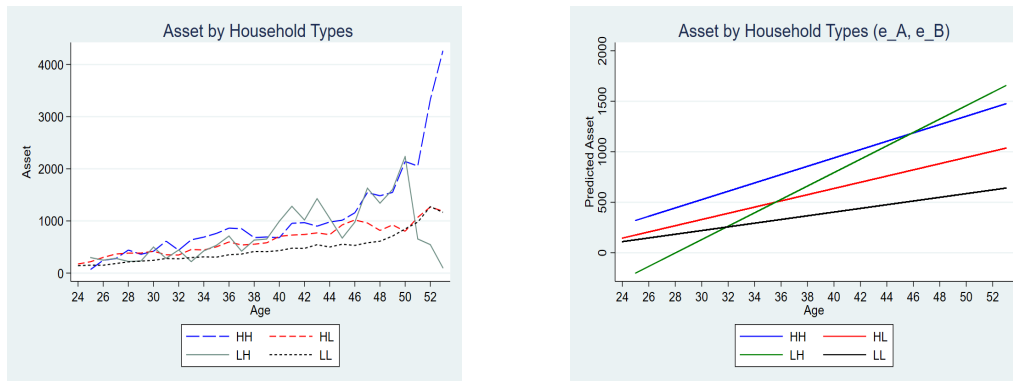


Figure 6. : Asset by Household Types

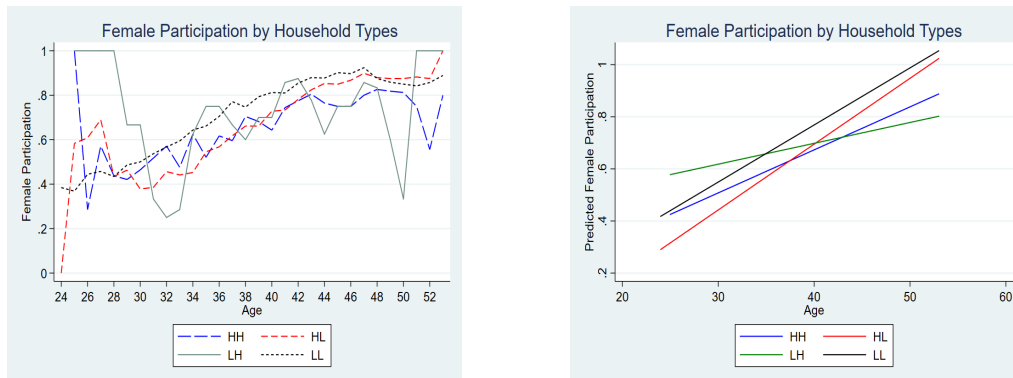


Figure 7. : Female Labor Force Participation by Household Types

B. Wage Process

The deterministic life cycle profile of male hourly wages is taken directly from the paper by Lise and Yamada (2019). They estimate this wage profile on JPSC data from 1993 to 2013. The age profile starts from age 24 and ends at age 53 for the benchmark cohort. However, Lise and Yamada (2019) report only a pooled wage profile for males and females. Because the life cycle earnings profile by human capital accumulation for females have different shapes compared to their male counterparts observed from the data, in the model I assume that deterministic life cycle wage profiles of female follow cubic spline. For example, in the JPSC data, average hourly wages of working college educated females is 0.507 times that of college educated males; I calibrate the level of female wages to match this earnings ratio.

Turning to the stochastic component of wages, the covariances matrix of permanent wage shocks is taken from Voena (2015), and modified in the calibration process so that it matches the life cycle mean of asset levels for the four household types, respectively. As in her work, the permanent wage shocks are assumed to be i.i.d. across time and uncorrelated between the two earners of a household:

$$\begin{bmatrix} \sigma_{\zeta_A}^2 & \sigma_{\zeta_B, \zeta_A} \\ \sigma_{\zeta_A, \zeta_B} & \sigma_{\zeta_B}^2 \end{bmatrix} = \begin{bmatrix} 0.15 & 0 \\ 0 & 0.1 \end{bmatrix}$$

C. Discount Factor and Interest Rate

I adopt the Voena (2015) choice of real risk-free interest rate of $r = 2$ percent per year. The discount factor, a key determinant of household precautionary (and life cycle) saving, is calibrated such that the average asset level for households aged 24 to 53 in the model ranges from 368.7 to 890.5 by household types, as measured in the JPSC dataset. This delivers a time discount factor of 0.99.

D. Income Tax

I refer to the 2003 income tax rates in Japan since the sample spans from 1993 to 2013. In Japan, each earner in the household is taxed individually based on which bracket their income level lies within. The tax regime follows a progressive system shown in figure 8.

To determine the income tax rate for each of the four types of individuals, I calculate the average of their life cycle annual earnings in 10,000 Japanese yen. The average annual earnings for college educated male, non-college educated male, college educated female, and non-college educated female are as follows: 1004.4, 666.5, 321.1, 238.0. As a result, the respective income tax rates they face are: 30%, 20%, 10%, 10%.

2003 National Income Tax Brackets (Japan)

Taxable Income Range (JPY)	Tax Rate	Deduction (JPY)
0 – 3,300,000	10%	0
3,300,001 – 9,000,000	20%	330,000
9,000,001 – 18,000,000	30%	1,230,000
Over 18,000,000	37%	2,700,000

Figure 8. : Income Tax Rates by Annual Earnings

E. Retirement Replacement Rate

The retirement replacement rate in Japan in 2003 refers to the ratio of an individual's pension benefits to their pre-retirement income. This measure is used to evaluate how effectively pensions replace earnings after retirement. According to OECD data on replacement rates in Japan (2003), the gross replacement rate is approximately 34-42% for an average worker. The model assumes each individual earns 40% of the pre-retirement income for each period during retirement.

F. Fix Utility Cost of Female Participation

In the data, only female workers have significant labor market nonparticipation rates. Since the JPSC sample is based on households of married wife with male members, I do not include an extensive margin of male labor supply decisions in the model. Based on empirical observation, the married female labor force participation depends more on the husband's education level than the education level of oneself. To generate empirically plausible female labor supply decisions along the extensive margin in the model, the fixed utility cost of married female labor market participation $q_{-e,m}$ is chosen such that the average female nonparticipation rate of households aged 24 to 53 is 34%, as in the JPSC dataset. Also observed in the data is the fact that all divorced females with a college degree choose to participate, at the same time, divorced females without college degrees have a nonparticipation rate of 13%. There are reflected in the choice of $q_{e,d}$.

G. Preference Parameters

With my preference specification, the parameter γ is the inverse of consumption Frisch elasticity, and set to be equal to 2. The number is adopted from Voena (2015) then adjusted upward to match the life cycle patterns of consumption and asset accumulation. I also directly adopted the parameter for economies of scale ρ in the household to capture the benefit of marriage compared with divorce to match the low divorce rate in the JPSC dataset.

H. The Pareto Weight

The husband's bargaining power at the time of the marriage, θ , is a function of the distribution factors known at $t = 1$. Information known at the start of the life cycle fixes the cross-sectional weight.

As the primary distribution factors known at the time of marriage in my baseline specification, I include the difference between the log wage fixed effect for the wife and husband, and log household asset at the time of marriage:

$$z_0 = \left\{ \ln(h_0^{A,e}) - \ln(h_0^{B,e}), \ln(a_0) \right\}'.$$

Bargaining power of the husband at the time of marriage:

$$\hat{\theta}_0 = \hat{\alpha} + \hat{\beta}' z_0.$$

In order to uncover the exogenous cross-sectional distribution of the Pareto weight $\hat{\theta}_0$, I run a Ordinary Least Squares regression using data from Japanese Panel Survey of Consumption that generates regression coefficients $\hat{\alpha}$ and $\hat{\beta}$. I use the private consumption share of the husband as a proxy for his Pareto weight because the model does not include other possible proxies such as leisure hours share, home hours share, market hours share, or public expenditure share.

$$\theta = \hat{\alpha} + \hat{\beta}_1 \cdot \ln\left(\frac{\bar{w}_A}{\bar{w}_B}\right) + \hat{\beta}_2 \ln \bar{a}$$

Where I regress the time-series average of husband's private consumption share on the time series average of log earnings gap between spouses, and on the time series average of the amount of household assets for each household in the sample. Based on the estimated results from Lise and Yamada (2019), the main source of dispersion in the Pareto weight comes at the time of marriage. In the cross section, the relative weight ranges between 0.22 and 2.1, with a mean of 0.8. I normalize the derived Pareto weights of the wife so that they have the matched mean of 0.44, and ranges between 0.2 and 0.7.

VI. Estimation Results

Table 5 summarizes the calibration of the model.

A. Model Fit

Table 6 panel A reports the exogenously calibrated empirical targets as well as the (near-perfect) fit of the model along these dimensions. Table 6 panel B reports endogenously calibrated empirical targets for both married couple and divorcee, along with fit of the model along these dimensions.

To further test the performance of the model, I examine the untargeted moments such as life-cycle earning profile of women and men for H and L types

Table 5—: Calibrated Model Parameters

Parameter	Governing	Value
Panel A: Consumption Preferences		
β	Discount factor	0.99
$q_{-L,m}$	Disutility of participation	0.0028
$q_{-H,m}$	Disutility of participation	0.0023
$q_{L,d}$	Disutility of participation	0.013
$q_{H,d}$	Disutility of participation	0
γ	Inverse Frisch elasticity	2
ρ	Economies of scale	0.3
Panel B: Marital Preferences		
σ_{ϵ_A}	Match quality shock	0.85
σ_{ϵ_B}	Match quality shock	1.05
μ_0	Marital compatibility	0.5
μ_1	Marital compatibility	1
$s^{A,e}$	Stigma cost of divorce	10, 30
$s^{B,e}$	Stigma cost of divorce	100, 300
Panel C: Financial Market		
r	Risk-free interest rate	0.02
$\tau^{j,e}$	Income tax rates	0.3, 0.2, 0.1, 0.1
Panel D: Wage Process		
$\frac{h_0^{B,e}}{h_0^{A,e}}$	Fixed-effect wage ratio	0.81, 0.79, 1, 0.97
$\sigma_{\zeta^A}^2$	Permanent shock	0.15
$\sigma_{\zeta^B}^2$	Permanent shock	0.1
$\sigma_{t_1,A}^2$	Initial dispersion	0.74
$\sigma_{t_1,B}^2$	Initial dispersion	0.32

respectively. Results are shown in figure 9 and 10.

I now report, the model-implied mean profiles over the life cycle of consumption, asset, and female labor force participation for married couple and divorcee by education level.

Figure 11 displays average consumption, asset, and labor supply profiles from age 24 to 80, for model generated results. As is common in life cycle models, assets rise over the life cycle as households accumulate wealth to fund retirement consumption (and because $r > \frac{1}{\beta} - 1$), but also for precautionary reasons to hedge against stochastic wage fluctuations. Assets in the model peak at retirement (age 65 in the model) and are then drawn down to fund retirement consumption. Note that in the model, by the start of the life cycle, households on average have

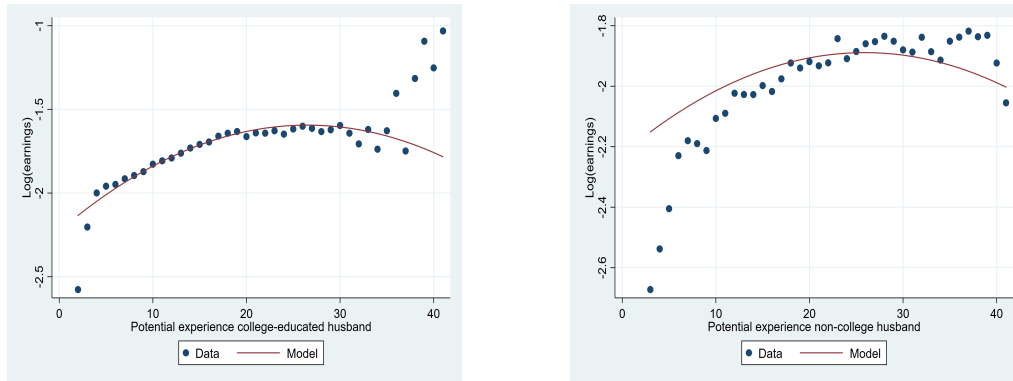


Figure 9. : Earnings Profile of Males

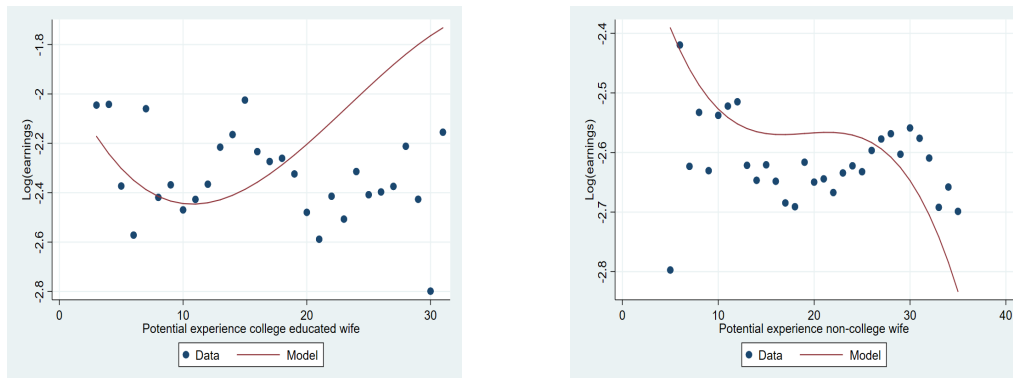


Figure 10. : Earnings Profile of Females

Table 6—: Empirical Targets Matched

Empirical targets	Data	Model
Panel A: Exogenously Calibrated		
College male labor income	-1.66	-1.66
Non-college male labor income	-2.06	-2.06
College female labor income	-2.34	-2.34
Non-college female labor income	-2.62	-2.62
Mean Pareto weight on wife	0.44	0.44
Panel B: Endogenously Calibrated		
Married couple consumption	13.6, 12.3	13.8, 12.7
	10.2, 11.1	10.6, 10.2
Married couple asset	890.5, 596.0	879.3, 593.0
	773.4, 368.7	767.9, 378.2
Married female participation	64.8%, 65.0%	64.5%, 64.6%
	66.2%, 66.4%	67.6%, 67.7%
Female divorcee participation	1, 86.6%	1, 87.3%
Divorce rate	0.23%	0.38%

accumulated considerable amount of assets (see panel b of Figure 11).

Consumption in the model rises over the life cycle until around age 50 in the model before starting to decline. In the earlier stage, wage and earnings risk, and the associated precautionary saving, as well as fairly high degree of patience ($r - (\frac{1}{\beta} - 1) \approx 1\%$) lead to low consumption early in life and subsequent positive consumption growth. Overall, the model captures well the growth of consumption and assets over the life cycle in the data. Panel (c) of Figure 11 shows average married female labor force participation rates throughout the life cycle. There is considerable non-participation with the lowest rate at 20%. An U-shaped pattern is observed, meaning that female participation decreases with age until 40, before increasing with age.

Now I analyze the life cycle patterns of married couples by their types for consumption, asset accumulation, and female labor force participation rate. Panel (a) shows the consumption profiles of married couple. *HH* type consumes the most, followed by *HL* type, *LH* type, and *LL* type. The ranking remains until retirement, and there are considerable consumption gaps between types. In Japan, husband is the primary earner, the type of the husband largely determines the consumption level of the household. However, the education type of wife determines when will the consumption level peak within a life cycle. For instance, household with college-educated wife continues to increase consumption until a later stage in life, while household with non-college educated wife decrease

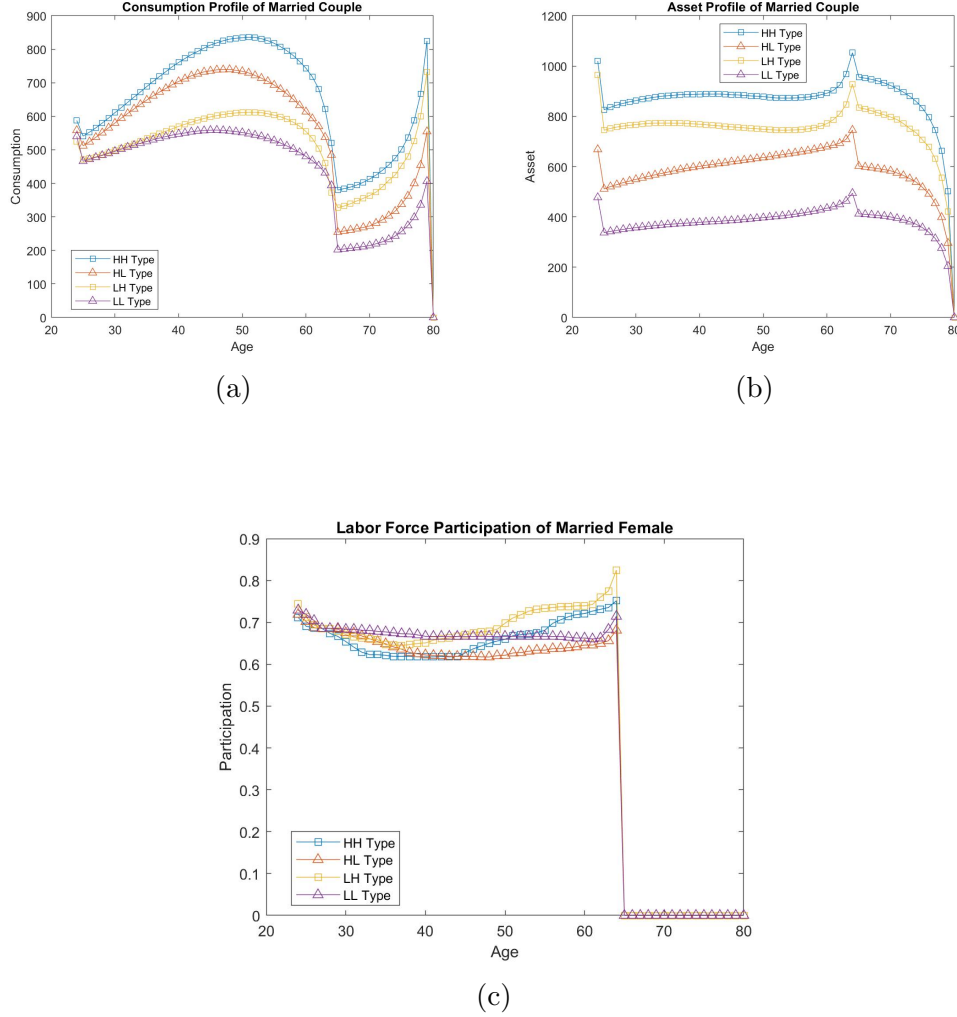


Figure 11. : Life Cycles of Cross-Sectional Means

Note:

This figure shows the life cycles of cross-sectional means in the benchmark model with additively separable preferences for married couples age 24-80. There are four types of household in each panel by the couples education level: *HH* type for college educated husband with college educated wife, *HL* type for college educated husband with non-college educated wife, *LH* type for non-college educated husband with college-educated wife, *LL* type for non-college educated husband with non-college educated wife.

their consumption well before retirement. After retirement, the consumption level rankings change in terms of the second and the third place. *LH* type surpasses *HL* type after age 65.

Panel (b) shows the asset accumulation profiles of married couple. *HH* type consumes the most, followed by *LH* type, *HL* type, and *LL* type. The ranking

remains until age 80, and there are considerable asset gaps between types. The household type that consumes the most also has the highest asset level, and the household type that consumes least also has the lowest asset level. While at the same time, *HL* type consumes more but accumulates less asset, *LH* type consumes less but accumulates more asset.

Panel (c) shows the female labor force participation profiles of married couple. Although unlike consumption or asset, there is no constant participation ranking throughout the working age, there are two pair of relationships that can be observed. If a college-educated female marries a non-college educated male, she will always have a higher probability of working compared to her counterpart who marries a college-educated male. At the same time, if a non-college educated female marries a non-college educated male, she will always have a higher probability of working compared to her counterpart who marries a college-educated male. Which means that married female labor force participation rate depends on the husband's education/income level, marrying a high type husband lowers the participation rate of married female controlling for the wife's type. Especially at the later stage of the life cycle (past 50), it becomes significant that college educated females have a higher probability of working compared with their non-college educated counterparts. The ranking in participation rate after 50 years old is: $LH > HH > LL > HL$.

Figure 12 displays model generated results of average asset level and labor supply profiles from age 24 to 80 for divorced individuals. As is common in life cycle models, assets rise over the life cycle as households accumulate wealth to fund retirement consumption (and because $r > \frac{1}{\beta} - 1$), but also for precautionary reasons to hedge against stochastic wage fluctuations. Assets in the model peak at retirement (age 65 in the model) and are then drawn down to fund retirement consumption. This trend of asset accumulation is commonly shared across different types of individuals except for college educated female. Their asset level first decrease relatively slowly until age 50 before rapidly increasing till retirement. Note that in the model, by the start of the life cycle, households on average have accumulated considerable amount of assets (see panel a and b of Figure 12).

Overall, the model captures well the growth of assets over the life cycle in the data. Panel (c) of Figure 12 shows average divorced female labor force participation rates throughout the life cycle. There is considerably lower non-participation compared with their married counterparts. College educated female always works throughout their life. An U-shaped pattern is observed between age 24 and 55 for non-college educated female divorcee, meaning that female participation decreases with age until 30, before increasing with age, and decreasing again around age 50.

Now I analyze the life cycle patterns of divorced individuals by their types for asset accumulation and female labor force participation rate. Panel (a) and Panel (b) show the asset accumulation profiles of female and male divorcee, respectively. Low type female accumulated more asset compared to their college-educated peers

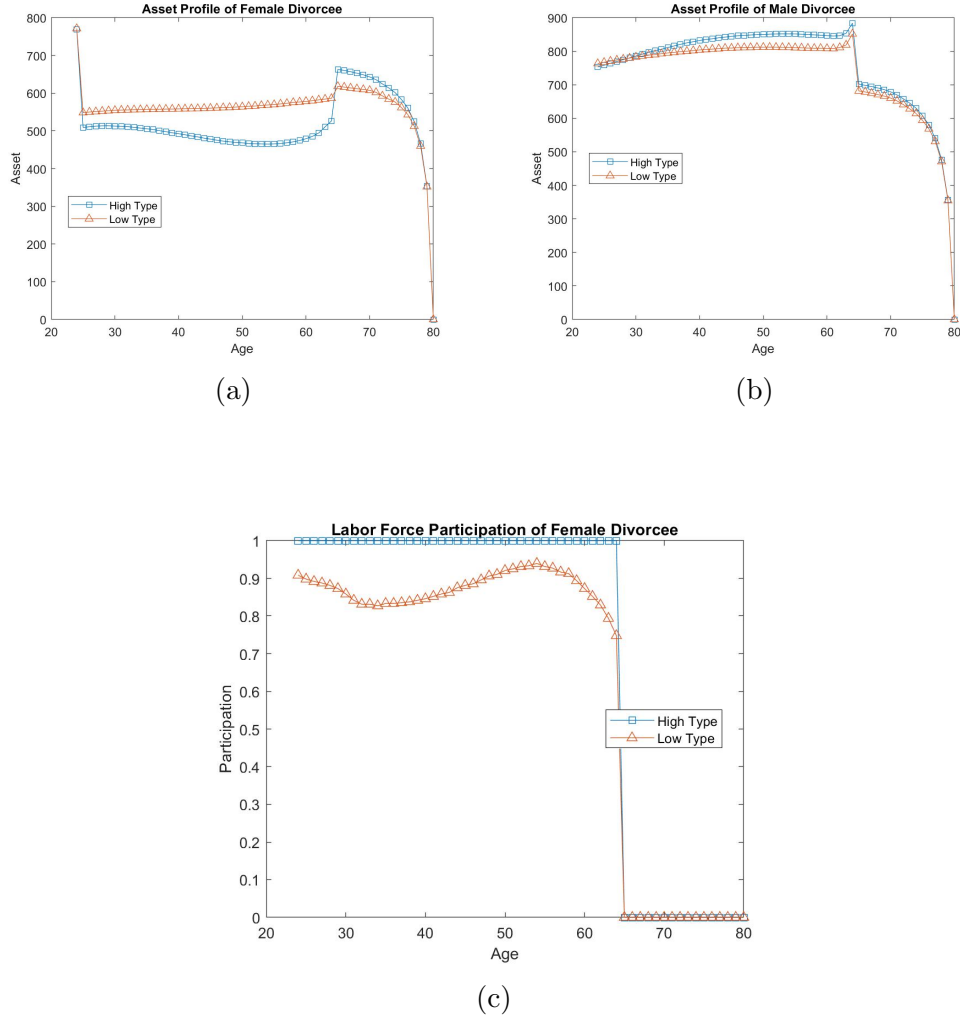


Figure 12. : Life Cycles of Cross-Sectional Means

Note: This figure shows the life cycles of cross-sectional means in the benchmark model with additively separable preferences for divorced individuals age 24-80. There are two types of household in each panel by the individual's education level: High type for college educated, Low type for non-college educated.

for every period before retirement. After age 65, asset level for high type female becomes constantly larger than that of low type female. Male divorcee of both types have significantly higher level of asset comparing with female divorcees. At the same time, high type male accumulates more asset for every period of the life cycle from the start until the end. However, there is less asset gap between types for male divorcee compared with female divorcee.

Panel (c) shows the female labor force participation profiles of divorcee. College

educated female divorcee has 100% participation as in the JPSC dataset. For non-college educated female divorcee, participation lowers to 80% at the start of age 30, then starts to increase until age 50 before decreasing again. The average participation rate for low type female divorcee is 87% in the benchmark model as in the dataset, showing a fairly well match of female labor force participation life cycle patterns between the benchmark model and the dataset.

B. The Pareto Weight and Female Participation

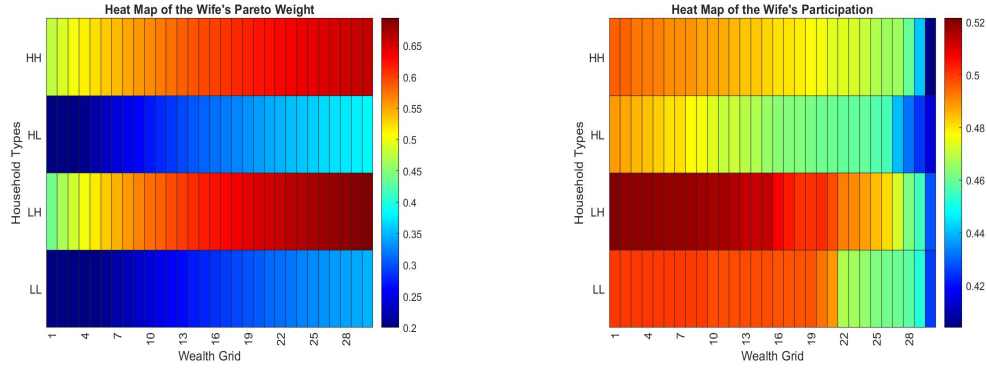


Figure 13. : The Pareto Weight and Female Participation

Note: This figure shows the cross-sectional variation of the Pareto weight and female labor force participation in the benchmark model with additively separable preferences for married couple age 24-80. There are four types of household in each panel by the combination of their education levels, there are 30 wealth grids with level of assets from low to high in each panel.

Now I analyze the cross-sectional patterns of married couples by their types for the Pareto weight and married female labor force participation rate. The left graph shows how the Pareto weight of wife varies with the household type and household asset level. It can be concluded that college educated wife has higher bargaining power comparing to their non-college educated counterparts. Especially, if a college-educated wife marries a non-college educated husband, the Pareto weight can be as high as 70%. On the contrary, if a non-college educated wife marries a college educated husband, her Pareto weight can be as low as 20%. Another pattern to be noticed is the low bargaining power of wife for *LL* type household, even if the spouses have the same education level, the resource allocation inside the household could be very unequal. The higher the asset level of the household, the higher the wife's bargaining power, regardless of the household type.

The right graph shows how the wife's labor force participation rate varies with household type and asset level. College educated wife marrying non-college educated husband has a significantly higher rate of participation compared with

other cases. It is partly due to their relatively higher income, and at the same time due to their ability to receive larger share of resources in the allocation stage since they have higher bargaining power. For the similar reason, but completely opposite case applies to non-college educated wife marrying college-educated husband, leading to the lowest rate of participation among the four types. It is worth noticing the high female labor force participation rate for *LL* type, even if the wife has the lowest bargaining power among different types of households, her probability to work is the second highest. More information is needed to discuss why this is the case and its policy implications. It would shed light on the topic since *LL* type consists of the largest share of households in the JPSC dataset. The higher the asset level of the household, the lower the wife's rate of participation, regardless of the household type.

VII. Concluding Remarks

In this paper, I show that intra-household bargaining matters for female labor force participation decisions. Starting from a simple non-unitary model with only two periods, the analytical results suggest that private consumption allocation is proportional to the Pareto weight. The female participation surplus in terms of additional life time value gained from working is inversely related to the bargaining power of wife. Extending this simple framework to a classical life cycle problem, I introduced heterogeneous households by education level, endogenous family composition transition, and cross-sectional variation of bargaining power. Results show that there are significant variation by types in terms of consumption and asset accumulation throughout the life cycle. There is also gender asymmetry in terms of asset accumulation patterns for divorced individuals. There is a general pattern that higher bargaining power of wife within the household leads to higher rate of participation. An exception is the household that consists of both non-college educated spouses. Despite the low bargaining power of wife, their rate of participation is the second highest.

In order to make use of the rich time use data, I plan to extend the utility function to include elastic working hours. This allows for the measurement of the second-earner effect in the intensive margin. I also plan to calibrate the model closely to the tax regime in Japan, conducting counterfactual analysis to the benchmark model that can shed light on policy responses to not only progressivity but also the collective vs. individual taxation debate. In addition to the future works mentioned above, I will do sensitivity analysis on the parametrization of the Pareto weight, and relaxing the non-renegotiation assumption to allow for dynamic adjustment of the bargaining power as new information is revealed when time evolves.

REFERENCES

- Attanasio, Orazio, Hamish Low, and Virginia Sanchez-Marcos.** 2008. "Explaining Changes in Female Labor Supply in a Life-Cycle Model." *American Economic Review* 98 (4): 1517-52.
- Blundell, Richard, Pierre-André Chiappori, Thierry Magnac, and Costas Meghir.** 2007. "Collective Labour Supply: Heterogeneity and Non-Participation." *Review of Economic Studies* 74 (2): 417-45.
- Blundell, Richard, Luigi Pistaferri, and Ian Preston.** 2008. "Consumption Inequality and Partial Insurance." *American Economic Review* 98 (5): 1887-1921.
- Chiappori, Pierre-André, Bernard Fortin, and Guy Lacroix.** 2002. "Marriage Market, Divorce Legislation, and Household Labor Supply." *Journal of Political Economy* 110 (1): 37-72.
- Choo, Eugene, Shannon Seitz, and Aloysius Siow.** 2008. "Marriage Matching, Risk Sharing and Spousal Labor Supplies." University of Toronto Department of Economics Working Paper 332.
- Cubeddu, Luis, and José-Victor Ríos-Rull.** 1997. "Marital Risk and Capital Accumulation." Federal Reserve Bank of Minneapolis Research Department Staff Report 235.
- De Nardi, Mariacristina, Eric French, and John B. Jones.** 2010. "Why Do the Elderly Save? The Role of Medical Expenses." *Journal of Political Economy* 118 (1): 39-75.
- Doepke, Matthias, and Michèle Tertilt.** 2009. "Women's Liberation: What's in It for Men?" *Quarterly Journal of Economics* 124 (4): 1541-91.
- Fernández-Villaverde, Jesús, and Dirk Krueger.** 2007. "Consumption over the Life Cycle: Facts from Consumer Expenditure Survey Data." *Review of Economic and Statistics* 89 (3): 552-65.
- Goldin, Claudia,** 2006. "The Quiet Revolution that Transformed Women's Employment, Education, and Family." *American Economic Review* 96 (2): 1-21.
- Gray, Jeffrey S.** 1998. "Divorce-Law Changes, Household Bargaining, and Married Women's Labor Supply." *American Economic Review* 88 (3): 628-42.
- Kocherlakota, Narayana R.** 1996. "Implications of Efficient Risk Sharing without Commitment." *Review of Economic Studies* 63 (4): 595-609.
- Ligon, Ethan.** 2002. "Dynamic Bargaining in Households (With an Application to Bangladesh)." Unpublished.
- Ligon, Ethan, Jonathan P. Thomas, and Tim Worrall.** 2000. "Mutual Insurance, Individual Savings, and Limited Commitment." *Review of Economic Dynamics* 3 (2): 216-46.

- Lise, Jeremy, and Shannon Seitz.** 2011. "Consumption Inequality and Intra-household Allocations." *Review of Economic Studies* 78 (1): 328-55.
- Low, Hamish, Costas Meghir, and Luigi Pistaferri.** 2010. "Wage Risk and Employment Risk over the Life Cycle." *American Economic Review* 100 (4): 1432-67.
- Lundberg, Shelly, and Robert A. Pollak.** 1993. "Separate Spheres Bargaining and the Marriage Market." *Journal of Political Economy* 101 (6): 988-1010.
- Mazzocco, Maurizio.** 2007. "Household Intertemporal Behaviour: A Collective Characterization and a Test of Commitment." *Review of Economic Studies* 74 (3): 857-95.
- Meghir, Costas, and Luigi Pistaferri.** 2004. "Income Variance Dynamics and Heterogeneity." *Econometrica* 72 (1): 1-32.
- Olivetti, Claudia.** 2006. "Changes in Women's Hours of Market Work: The Role of Returns to Experience." *Review of Economics Dynamics* 9 (4): 557-87.
- Peters, H. Elizabeth.** 1992. "Marriage and Divorce: Reply." *American Economic Review* 82 (3): 686-93.
- Voena, Alessandra.** 2015. "Yours, Mine, and Ours: Do Divorce Laws Affect the Intertemporal Behavior of Married Couples?: Dataset." *American Economic Review*. <http://dx.doi.org/10.1257/aer.20120234>.